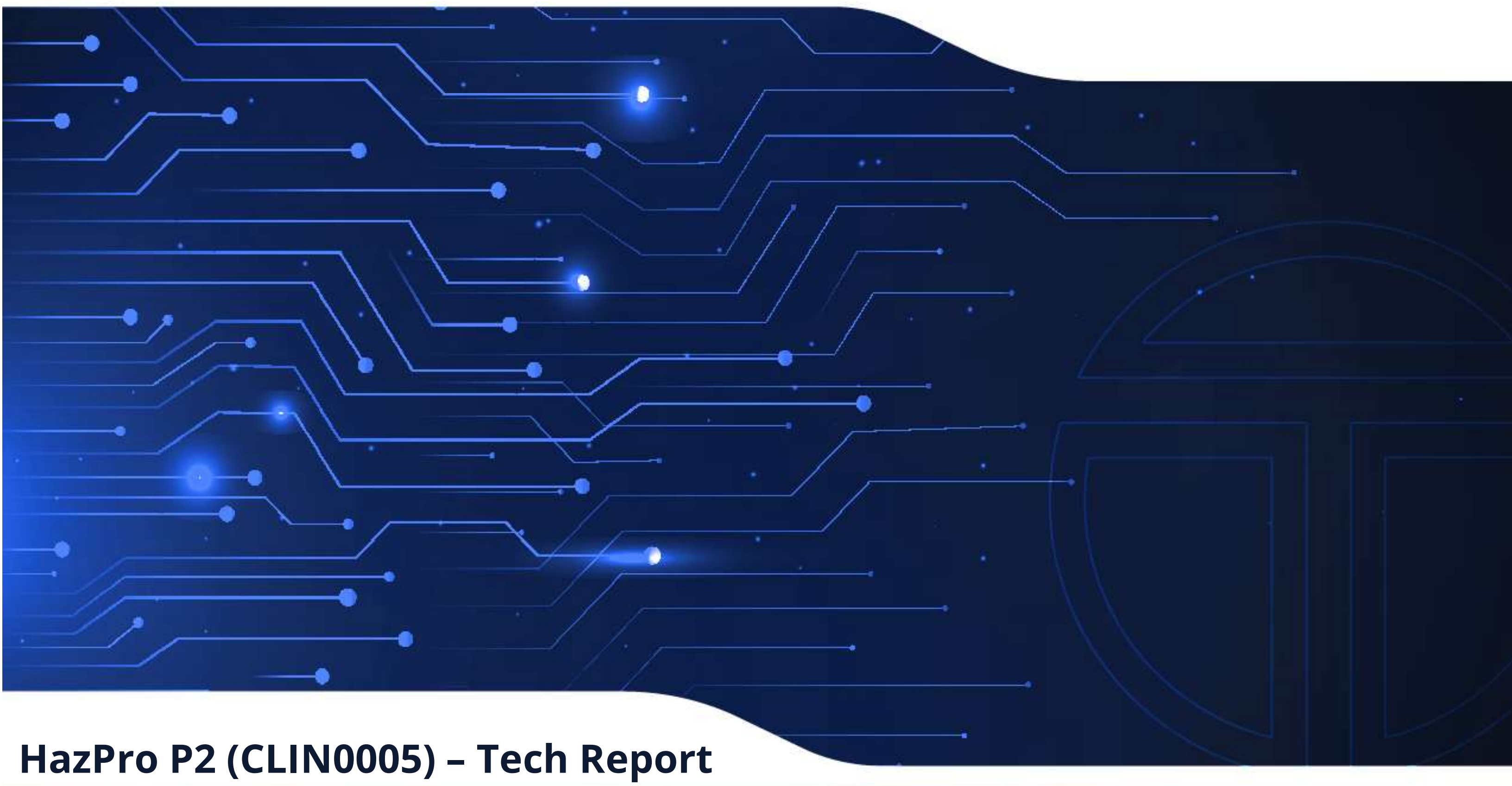

HazPro P2 (CLIN0005) – Tech Report

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Technical Report

Overview: This report documents the completion of the digital mobile application workflows associated with the preparation of hazardous materials for military air shipment, in accordance with the requirements outlined in AFMAN 24-604, Preparing Hazardous Materials for Military Air Shipments. This effort was executed as part of a Phase II prototype focused on digitizing and operationalizing hazardous materials workflows to support DoD logistics, deployment, and transportation operations.

The mobile application enables personnel—particularly Hazardous Materials Preparers and Inspectors—to perform their roles efficiently and compliantly through guided digital workflows, mobile accessibility, and integrated calculation and reference tools.

Project Scope: To develop HazPro mobile software Minimum Viable Product (MVP) to modernize the process for preparing and inspecting HAZMAT shipments for SDDGs based upon DoD regulations.

Assumptions: Customers want a scalable solution that supports Preparer and Inspector functions and allows quicker and effective Hazardous Materials shipment capabilities.

CLIN	Deliverable Description
0005	Develop & Demonstrate Preparer Software Features

Deliverable Focus: The scope for this CLIN deliverable included the design and development of the mobile application workflows that guide users through the end-to-end process of preparing hazardous materials for air shipment. These workflows account for hazard class identification, packaging instructions, documentation requirements, and special handling procedures, all aligned with **AFMAN 24-604**.

In addition to standard workflows, dedicated modules were implemented to support:

- Deviations, Waivers, and Special Requirements, allowing users to input applicable authorizations or required conditions under regulatory exception.
- Quick-access calculators and tools supporting operational decision-making in field conditions.

Accomplishments/Results:

Hazard Classes Addressed:

Workflows were completed for the following UN Hazard Classes:

- Class 1 – Explosives
  - Standard shipments
  - Grandfathered shipments (legacy packaging/documentation cases)
- Class 2 – Gases
- Class 3 – Flammable Liquids
- Class 4 – Flammable Solids, Spontaneously Combustible Materials, and Dangerous When Wet
- Class 5 – Oxidizers

- Class 6 – Toxic Substances and Infectious Substances
- Class 8 – Corrosive Materials
- Class 9 – Miscellaneous Hazardous Materials

Class 9 presented unique challenges due to the wide variation in item types, including lithium batteries, magnetized materials, and environmentally hazardous substances. Workflows were tailored accordingly to address these specialized requirements.

Organic Peroxides, though part of Class 5, were **not included** in this delivery due to their unique regulatory handling, temperature control requirements, and additional reference materials outside of AFMAN 24-604. These were deemed outside the scope of this Phase II effort.

In addition to class-specific workflows, the following specialized workflows were completed:

- **Deviations and Waivers**: Structured processes to ensure regulatory exception handling with proper documentation.
- **Special Requirements**: Embedded guidance for items with restricted handling, limited quantity provisions, or specific command stipulations.

Excluded Hazard Classes / Materials

The following were **excluded** from the scope of this deliverable due to complexity and alignment with external regulatory frameworks:

- Class 7 – Radioactive Materials
- Class 5 – Organic Peroxides only **(Note:** Oxidizers were completed**)**

These materials involve specialized packaging, temperature controls, or references to external federal agencies such as the NRC or IAEA, which were not within the scope of this phase.

Decision Support Tools

To enhance operational usability and reduce preparation time, a suite of quick-access decision support tools and calculators were integrated into the mobile application. These tools are accessible on the main application menu and throughout the workflow process and support field users in making accurate, compliant decisions aligned with regulatory guidance:

- **Compressed Gas Quantity Calculator** – Assists in determining shipping quantities based on pressure and cylinder volume.
- **Dry Ice Quantity and Aircraft Suitability Calculator** – Validates dry ice limits against aircraft capabilities.
- **Measurement Conversion Tool** – Supports metric-to-standard and standard-to-metric conversions for weight, volume, and linear measurements.
- **Placarding Requirements Tool** – Dynamically recommends placards based on hazard class and quantity thresholds.

In addition to these user-facing tools, the backend logic and rule structure for Hazardous Materials Compatibility and Segregation requirements is also completed. This capability ensures that materials are not loaded or stored in ways that violate compatibility tables or increase safety risks. The logic adheres to AFMAN 24-604 standards and includes:

- Class compatibility matrix enforcement
- Inter-class segregation rules
- Exceptions and special provisions for mixed loads

While this logic is fully implemented and validated, the frontend user interface (UI) for this capability will be integrated in the next CLIN deliverable, focused on Hazardous Materials Inspection workflows.

Key Outcomes for Hazardous Materials Preparers

The mobile application delivers two critical outputs at the end of each workflow:

1. **Package Marking and Labeling Requirements** – Automatically generated, shipment-specific marking and labeling guidance based on hazard class, packaging, and quantity.
2. **Completed Shipper's Declaration for Dangerous Goods (SDDG)** – A digital, compliant version of the required shipping declaration, ready for print or secure digital transmission.

These outputs reduce administrative overhead, ensure standardization, and help enforce proper documentation protocols across the enterprise.

Additional Complexity Areas

The following workflow challenges were successfully addressed during this period:

- **Shipments without Packing Groups (PG)** – Created custom workflow logic to guide users through preparation in the absence of defined PGs.
- **Materials without Performance-Oriented Packaging (POP)** – Developed alternative workflow logic for handling such materials through compliant means.
- **Grandfathered Materials** – Embedded decision branches for legacy labeling/marking while maintaining modern compliance standards.

Outcomes and Benefits

- Digitally guided workflows covering major hazard classes and special preparation cases.
- Standardization of preparation processes across units and personnel roles.
- Rapid generation of shipment documentation, improving timeliness and compliance.
- Reducing errors in packaging, labeling, and placarding through integrated decision tools.
- Increased confidence for field users through intuitive UI and embedded regulatory logic.

Demonstration

Demonstration of the completed Preparer features was conducted on 12 June 2025 for end users and key stakeholders. This demonstration showcased the digital workflows developed under this CLIN, including hazard class-specific preparation, deviations and waivers handling, decision support tools, and the generation of package labeling guidance and the Shipper's Declaration for Dangerous Goods (SDDG).

Stakeholders were provided a walk through of the application from the perspective of a Hazardous Materials Preparer and see how the workflows promote compliance with AFMAN 24-604 while streamlining operational tasks. Feedback gathered during the session was positive and will inform future phases of development, including upcoming work in the Hazardous Materials Inspection module.

Future Considerations

For full operationalization across all mission profiles, future development CLIN and/or phases may include:

- Workflow support for Organic Peroxides and Class 7 Radioactive Materials.
- Interoperability with multimodal regulatory frameworks (e.g., IATA, IMDG, 49 CFR).
- Integration with shipment tracking systems and deployment planning tools.

Targets for Next Deliverable:

CLIN	Deliverable Description
0006	Develop & Demonstrate Inspector Software Features